oncoming vehicles is restricted. It is also used in blind spots in shops.

DO YOU KNOW?

In the rear view mirror, the following sentence is written. "Objects in the mirror are closer than they appear". Why?

Problem 3

A car is fitted with a convex mirror of focal length 20 cm. Another car is 6 m away from the first car. Find the position of the second car as seen in the mirror of the first. What is the size of the image if the second car is 2 m broad and 1.6 m high?

Solution:

Focal length = 20 cm (convex mirror)

Object distance = –6m = –600 cm

Image distance v = ?

$$\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$$

$$\frac{1}{20}=\frac{1}{-600}+\frac{1}{v}$$

$$\frac{1}{v}=\frac{1}{20}-\frac{1}{-600}=\frac{1}{20}+\frac{1}{600}\ \frac{1}{v}=\frac{30+1}{600}=\frac{31}{600}$$

$$v=\frac{600}{31}=19.35\text{ cm}$$

b) Size of the image

$$m=\frac{-v}{u}=-\frac{v}{(-u)}=-\frac{600}{31}\times\frac{1}{-600}=\frac{1}{31}$$

$$\text{Breadth of image}=\frac{1}{31}\times 200\text{ cm}=6.45\text{ cm}$$

$$\text{Height of image}=\frac{1}{31}\times 160\text{ cm}=5.16\text{ cm}$$

6.6 Speed of light

In early seventeenth century, the Italian scientist Galileo Galilee (1564-1642) tried to measure the speed of light.

In 1665, the Danish astronomer Ole Roemer first estimated the speed of light by observing one of the twelve moons of the planet Jupiter. He estimated the speed of light to be about 220,000 km per second.

DO YOU KNOW?

Some organisms can make their own light too? This ability is called bioluminescence. Worms, fish, squid, starfish and some other organisms that live in the dark sea habitat glow or flash light to scare off predators.

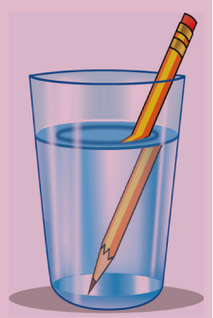
In 1849, the first land based estimate was made by Armand Fizeau. Today the speed of light in vacuum is known to be almost exactly 300,000 km per second.

6.7 Refraction of light

Activity 4

Refraction of light at air – water interface

Put a straight pencil into a tank of water or beaker of water at an angle of 45° and look at it from one side and above. How does the pencil look now? The pencil appears to be bent at the surface of water.



This activity explains the refraction of light. The bending of light rays when they pass obliquely from one medium to another medium is called refraction of light. Light rays get deviated from their original path while entering from one transparent medium to another medium of different optical density. This deviation (change in direction) in the path of light is due to the change in velocity of light in the different medium. The velocity of light depends on the nature of the medium

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